Setting up a Proxy *for* Docker Containers *in* Linux Machines

With the popularity of cloud computing on the rise, there has been a steady increase in the use of container technology by enterprises. Docker is a popular container technology that we will explore in this article. The setting up of a proxy to run Docker containers may prove daunting to some, so this article walks the reader through the process and elucidates the pitfalls.

Nowadays, everyone in the community knows what a Docker container is and many of the more technically inclined appreciate this technology. Gone are the days of monolithic and 'single point of failure' kind of applications. We're now in the era of microservices based applications, and Docker containers facilitate this with ease.

As of the end of 2018, container technology has been widely adopted as compared to VM (virtual machines in cloud), and this trend has been growing since early 2017.

It comes as no surprise that the most widely adopted container technologies are Docker (52 per cent) and Kubernetes (30 per cent); these two are leading the pack currently. Having said this, more than 70 per cent of the adopters (of Docker and Kubernetes) mentioned that they have deployed containers on VMs, so it's not likely that VMs are going to disappear. They're not, but going forward, IT executives do want to cut their VM licensing costs.

Containers are primarily being used for cloud-native applications (54 per cent). This is followed by lightweight stateless applications (39 per cent), cloud migrations (32 per cent) and modernising legacy applications (*https://www.zdnet.com/article/container-adoption-speeds-up-to-the-detriment-of-vms/*).

Proxies and the proxy server

A proxy server sits between a client application, such as a Web browser and a real server. It intercepts all requests to the real server to see if it can fulfil the requests itself. If not, it forwards the request to the real server.

In the Docker container environment too, proxy is very important and is used to handle connections originating from the local machine, which might otherwise not pass through the iptables rules that Docker configures to handle port forwarding or when Docker has been configured such that it does not manipulate iptables at all.

Setting up a proxy has become a major issue in many Docker container configurations nowadays because of the platform version differences, many applications getting containerised, and the availability of limited or scattered information/material, which is creating confusion.

As mentioned earlier, Docker container adoption and usage is growing every day and, often, people face many challenges while configuring proxies in a Docker container environment. Let us now look at how to address many of these challenges by providing the appropriate solutions and steps to be followed.

The Docker and Ubuntu versions used for this article are:

- Docker client 18.03.1-ce Community edition
- Docker server 18.03.1-ce Community edition
- Ubuntu 18.04.1 LTS Bionic release

On host proxy setup

This is a generic proxy setup for host/bare-metal servers to work on normal Internet connections.

1) *Problem*: *apt-get updates* and *install* will not work behind a proxy. When *$ sudo apt-get* is executed, an error '407 Proxy Authentication is required' is thrown up.